

Analysis III
Home Work 2

Student: 26-712-224

October 1, 2026

Exercise 1 (Step functions, 1+2+1 P.)

Let us define, for $x \in [0; 1]$,

$$f(x) = \sum_{k=0}^{+\infty} \chi_{[2^{-k-1}, 2^{-k}]}(x).$$

(a) Is f a step function? Justify your answer briefly (you do not need to give a complete proof).

(b) Show that there exists a step function $g : [0; 1] \rightarrow \mathbb{R}$ with $f = g$ almost everywhere and explicitly write down the set

$$\{x \in [0; 1] : f(x) \neq g(x)\}.$$

(c) Compute $\int_0^1 f(x) dx$ by making use of (b).

Answer:

(a) As it is given $f(x)$ is NOT a step function. The reason being that a step function is defined to consist of a finite number of intervals - the sum shown is not finite. Also, there is no finite interval representation for that exact same f .

(b) The simple step function: $g : [0, 1] \rightarrow \mathbb{R}$ given by: $g(x) = \chi_{[0, 1]}(x)$ is such that $f = g$ (a.e.).

Now the ends of the each interval: $\chi_{[2^{-k-1}, 2^{-k}]}$ are "double counted" - except for $k = 0$ when $\chi_{[2^{-k-1}, 2^{-k}]} = \chi_{[\frac{1}{2}, 1]}$.

Consequently, the points where $f \neq g$ is the set:

$$D = \left\{ \frac{1}{2^k} \mid k \in \mathbb{N}_+ \right\} \cup \{\emptyset\}$$

Not that D is countable and hence a set of measure zero. I.e. $f = g$ (a.e.).

(c) It is clear that g is Lebesgue Integrable.

Using Theorem 11.2.15 in the notes we can conclude that, for Lebesgue Integration:

$$\int g = \int f$$

As $\int g = \lambda([0, 1]) = 1$, we can conclude that:

$$\int f = 1$$

Exercise 2 (Sequences of step functions, 2+1+2+1 P.)

Let I be a compact interval in $\mathbb{R}^n$.

- (a) Let $f : I^\circ \rightarrow \mathbb{R}$ be continuous and defined on the interior I° of I . Assuming that f is unbounded from below, show that there cannot exist an increasing sequence $(f_k)_{k \in \mathbb{N}} \in T^{\text{inc}}$ of step functions with $f_k \rightarrow f$ almost everywhere.
- (b) Let $(f_k)_{k \in \mathbb{N}} \in T^{\text{dec}}$ be a decreasing sequence of step functions for which $f_k \rightarrow \chi_I$ almost everywhere. Show that

$$\int_I f_k(x) dx \rightarrow \lambda_n(I)$$

by reducing the problem to an application of Theorem 11.2.10(1).

- (c) Consider the sequence of step functions $f_k : [0; 1] \rightarrow \mathbb{R}$ given by

$$f_k(x) = k\chi_{[0; 1/k]}(x).$$

Prove that $f_k \rightarrow 0$ pointwise almost everywhere and compute $\int_0^1 f_k(x) dx$ for every $k \in \mathbb{N}$. Why is this not in contradiction with Theorem 11.2.10(1)?

- (d) Let $f, g : I \rightarrow \mathbb{R}$ be two functions so that $f = g$ almost everywhere. Show that if $f \in \mathcal{L}^{\text{inc}}$, then also $g \in \mathcal{L}^{\text{inc}}$ and

$$\int_I f(x) dx = \int_I g(x) dx.$$